

Yilin “Harry” Lu

Portfolio: <https://sites.google.com/umn.edu/yilinharry-lu/bio>

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EDUCATION

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- **University of Minnesota, Twin Cities** Minneapolis, MN
Bachelor of Arts with Honors; GPA: 3.8/4.0 August 2023 - December 2026
 - **Majors & Minor:** Triple Majors in Computer Science, Mathematics, Quantitative Economics (Minor: Statistics).
 - **Honors:** Undergraduate Research Scholarship (\$ 1,800), Top 1 Paper at the Economic Writing Competition (\$ 750), University Honor Program, Dean’s List (2025, 2024, 2023).
 - **Relevant Coursework:** Machine Learning, Visualization for Interpretable AI, Algorithm Design& Analysis, Data Structures, Discrete Mathematics, Applied Linear Algebra, Bayesian Statistics for Astrophysics, Web Infrastructure.
 - **Cornell University** Ithaca, NY
Visting Student in Economics; GPA: 4.3/4.3 Winter 2025
 - **San Diego State University** San Diego, CA
Concurrent Enrollment in Mathematics while in the High School 2022 - 2023

RESEARCH EXPERIENCE

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- **University of Pennsylvania** Philadelphia, PA
Undergraduate Research Intern - Fully-funded (Human Computer Interaction, AI4Science) May 2026 - Present
 - **Research Area(s):** Tools of Thoughts for Scientific Reasoning.
 - **College of William & Mary** Williamsburg, VA
Undergraduate Research Intern (Natural Language Processing, AI4Science) Nov 2025 - Present
 - **Research Area(s):** Autonomous scientific research [P7], agentic memory for scientific discovery.
 - **University of Minnesota, Twin Cities** Minneapolis, MN
Undergraduate Research Lead (Explainable AI, Human Computer Interaction, Visualization) Jan 2024 - Present
 - **Research Area(s):** Explainable AI for non-expert users [P3-P6, T1-T5], AI agents and their applications [P2, P8].
 - **Minnesota Institute for Astrophysics** Minneapolis, MN
Undergraduate Research Intern (Human Computer Interaction, AI4Science) September 2023 - May 2024
 - **Research Area(s):** Crowdsourcing systems for scientific discovery [P1], image processing for transient detections.

PUBLICATIONS

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- [P8] **Macro and Micro Uncertainty Signals for Prefix-Level Risk Prediction in Web Agents:** *In Submission to The Conference of Language Modeling (COLM 2026)*. S. Pan, Y. Shen, **Y. Lu**, Q. Wang.
 - [P7] **SciDER: Scientific Data-centric End-to-end Researcher:** *In Submission to ACL 2026 (Demo Track)*. K. Lin*, **Y. Lu***, S. Bhat, X. Guo, J. Oliver, Q. Wang. (* = denoted core contributors.)
 - [P6] **Play, Reflect, Understand: Developing a Web-Based Game for Learning Multi-Agent LLM Systems:** *In submission to ACM User Interface Software and Technology (UIST 2026)*. **Y. Lu**, S. Du, Q. Wang.
 - [P5] **GNN101: Visual Learning of Graph Neural Networks in Your Web Browser:** *IEEE Transactions on Visualization and Computer Graphics (TVCG)*. **Y. Lu**, C. Chen, Y. Chen, K. Huang, M. Zitnik, Q. Wang.
 - [P4] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *IEEE Visualization Conference (VIS 2025)*. **Y. Lu**, S. Du, Q. Wang.
 - [P3] **What Can a Node Learn from Its Neighbors in Graph Neural Networks:** *IEEE Visualization Conference (VIS 2024), 7th VISxAI Workshop*. **Y. Lu**, C. Chen, M. Xu, Q. Wang.
 - [P2] **What Helps Beyond Semantic Similarity? Utility-Driven Memory Selection for Robust Large Language Model Function Calling:** *Manuscript under revision*. J. You, M. Gini, W. Zhu, S. Chen, **Y. Lu**, Y. Mao.
 - [P1] **GRB231115A: TURBO Pre-Burst Optical Upper Limits:** *NASA General Coordinates Network (NASA GCN Circular)*. R. Strausbaugh et al., including **Y. Lu** (6th author).

PRESENTATIONS

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- [T6] **Growth Pattern Analytics of Digital Infrastructures:** *UMN Undergraduate Research Workshop*. **Y. Lu**.
 - [T5] **GNN 101: Visual Learning of Graph Neural Networks in Your Web Browser:** *IEEE VIS 2026*. **Y. Lu**.
 - [T4] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *IEEE VIS 2025*. **Y. Lu**.
 - [T3] **The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay:** *UMN CS&E Graduate Student Orientation Poster Session (Invited)*. **Y. Lu**, S. Du.
 - [T2] **GNN101: Visual Learning of Graph Neural Networks in Your Web Browser:** *UMN Visual Computing Seminar (Invited)*. **Y. Lu**.
 - [T1] **What Can a Node Learn from Its Neighbors in Graph Neural Networks:** *IEEE VIS 2024*. **Y. Lu**.

AUTHORED PACKAGES

- **[L3] kandor: An LLM-based World Building Engine:** A modularized LLM-based world building engine with a temporal knowledge graph memory system and an interactive visualization system for exploration in Python.
- **[L2] LargeGraphs.jl.: Large Graph Sensemaking in Julia:** A webGL-based library (powered by Sigma.js) that makes large graphs explorable and explainable through interactive visualization in Julia.
- **[L1] gnn-exp: An Python Package for Visualizing Graph Neural Networks:** A package to visualizing different types of graph neural network for neuron-level interpretability and explainability.

ACADEMIC SERVICE

- **IEEE Trans. Visualization and Computer Graphics:** Invited Student Reviewer (2025 - Present).

PRESS

- **College of William & Mary pioneers AI scientist: SciDER enables machines to independently carry out the entire research process (in Mandarin):** *TechWalker*.

COMPUTER SCIENCE RESEARCH PROJECTS

- **SciDER: Scientific Data-centric End-to-end Researcher [90 stars]:** A multi-agent system with a data processing agent, an experiment agent, and an ideation agent with self-evolve memory bank system, integrated in a Python package.
- **Agentopia: A Gamification Framework for Understanding Hallucination in LLMAS [5 stars]:** A customizable and grammar-driven game framework for understand and experiment hallucination in large language multi- agent systems.
- **ScholaVIS: A Toolkit for Intent Alignment Analysis of Human-AI Co-writing [5 stars]:** A scientific writing simulation tool integrated with different LLM agents, data analysis pipeline for event study, with a visual analytic software .
- **GNN101: An Interactive Visualization Tool for Graph Neural Networks [36 stars]:** An interactive tool for graph neural networks, developed in collaboration with Stanford and Harvard Medical School, deployed at Stanford CS224W and gained 1.6k+ users, designed a comparative study to evaluate the tool effectiveness against a baseline.

MATHEMATICS AND ECONOMICS RESEARCH PROJECTS

- **Extending Partial Orders: From Binary Relations to Linear Orderings:** A short note on ordering theory and set theory, specifically about how to extend binary relations to linear orderings.
- **Digital Drifting: Growth Pattern Analytics of Digital Infrastructures in the US: (Top 1 Project in the Econ Department)** A spatial-temporal convergence analysis of digital development in the US with a RShiny visualization.

RESEARCH MENTORSHIP

- **Shurui Du:** *Bachelor of Science in Computer Science, University of Minnesota.* Co-authored Works: [P4, P6, T3].
- **Yuxin Chen:** *Bachelor of Science in Computer Science, University of Minnesota.* Co-authored Work: [P5].
- **Chongwei Chen:** *Bachelor of Arts with Honors in Mathematics, University of Minnesota.* Co-authored Works: [P3, P5].
- **Matthew Xu:** *Bachelor of Science in Computer Science& Philosophy, UIUC.* Co-authored Work: [P3].

SKILLS SUMMARY

- **Languages:** Python, C/C++, TypeScript, JavaScript, Julia, R.
- **Visualization Technologies:** Unity 3D, D3.js, React.js, Phaser.js, IPython& IJulia, Flask, Streamlit, RShiny.
- **Machine Learning & Data Science:** Numpy, Pandas, Matplotlib, PyTorch, LangGraph, Scikit-learn, Transformers.
- **User Study:** Statistical Analysis, Experiment Design, Quantitative/Qualitative Analysis, Crowdsourcing.
- **Infrastructure& Deployment:** Serverless (Vercel, GitHub Pages), GPU Programming, AWS Deployment, Docker.

REFEREES

- **Qianwen Wang:** *Assistant Professor in Computer Science, University of Minnesota.* Email: qianwen@umn.edu
- **Qingyun Wang:** *Assistant Professor in Data Science, William& Mary.* Email: qwang16@wm.edu.
- **Andrew Head:** *Assistant Professor in Computer Science, University of Pennsylvania.* Email: head@seas.upenn.edu.